

# What Does a Sim-to-Real Correlation Support?

## A Bounded 26-Paper Reporting Audit and Decision-Level Reanalysis

Tri Lam - 6 August 2026

### Abstract

Sim-to-real correlation is often used to support robot-policy evaluation, but the practical question is whether acting on an evaluator selects an acceptable real policy. A coefficient describes association among its entering points; it does not specify the action, loss, or transport to new policies and tasks. We audit 26 papers and find finite displayed panels in all 26; none defines a target population or probability-sampling mechanism.

In this non-systematic set, top-1 decisions agree in 17 of 19 complete matrices. Real2Sim’s checkpoint-selection aggregate has  $r = .878-.947$  but selects a different policy, with 12.5 percentage points of displayed real regret. Equal task weighting for Cosmos-Surg gives  $r = .883$ , Spearman  $\rho = .371$ , a different winner, and 10.0 points of regret. OSCAR’s plotted values give  $r = .855$  with a 1.1-point argmax difference, but its axes come from different evaluation pools and its uncertainty cannot be reconstructed. Rank-aware companion metrics add useful evidence but do not specify the downstream action, acceptable loss, or real-test budget.

High correlation can coexist with task disagreement, reversed checkpoint selection, and metric-implementation discrepancies. Correlation alone does not determine winner agreement or regret.

### 1. Introduction

Real-robot evaluation is expensive, so simulation, learned world models, and video evaluators are increasingly used to rank policies before deployment. The consequential act is usually not reporting a coefficient; it is retaining, testing, or deploying a policy because of that coefficient. Correlation between simulated and real performance is relevant to that use, but three different claims are often left entangled:

1. simulated and real scores are associated for the displayed points;
2. that association extends to new policies, tasks, sites, or sessions; and
3. selecting by simulation produces an acceptable real-world decision.

The first claim is descriptive. The second requires a target population and a design or model linking the observed panel to that population. The third requires a decision rule and loss. A high coefficient can coexist with a ranking reversal near the top because Pearson correlation summarizes average linear association rather than winner identity or decision regret.

Prior work has examined sim-to-real predictivity, repeated-observation correlation, and reliable reinforcement-learning evaluation [1-5]. Recent robotics systems have made simulation-based evaluation more capable and reproducible [6-10]. Policy selection under uncertainty, top-k regret, and downstream regret are also established ideas [33-35]. We apply them to ask:

What does a reported sim-to-real coefficient support for a specified target, selection rule, and loss?

We contribute:

- a framework separating association, finite-panel action, population transport, and decision acceptability;
- an audit of 26 papers’ correlation claims and a 19-case decision ledger;
- task-composition and measurement sensitivities;
- a reconstruction of six Real2Sim MMRV values; and
- a reporting checklist for future evaluations.

## 2. Association, population, and decision

Let  $Y_i^R$  and  $Y_i^S$  be real and simulated scores for displayed unit  $i$  in panel  $\mathcal{P}$ . The Pearson coefficient

$$r_{\mathcal{P}} = \frac{\sum_{i \in \mathcal{P}} (Y_i^R - \bar{Y}^R)(Y_i^S - \bar{Y}^S)}{\sqrt{\sum_{i \in \mathcal{P}} (Y_i^R - \bar{Y}^R)^2} \sqrt{\sum_{i \in \mathcal{P}} (Y_i^S - \bar{Y}^S)^2}}$$

is a property of that displayed panel. The index  $i$  may represent policies, checkpoints, tasks, conditions, or a product of axes. A population correlation requires a target distribution over the relevant units and a defensible link from the observed panel to that distribution.

A policy-selection decision requires different quantities. For candidate set  $\mathcal{A}$ , define the displayed real and simulated winner sets as

$$R = \arg \max_{a \in \mathcal{A}} Y_a^R, \quad S = \arg \max_{a \in \mathcal{A}} Y_a^S.$$

Selection is possibly correct when  $R \cap S \neq \emptyset$ , and robustly correct when every simulated maximizer is a real maximizer,  $S \subseteq R$ . Displayed real regret for selected policy  $a$  is

$$\max_{b \in \mathcal{A}} Y_b^R - Y_a^R.$$

These definitions preserve ties and expose the decision consequence. They do not convert a convenience panel into a policy population.

WorldGym illustrates why the axes cannot be collapsed into one effective sample size. Its Figure 4a contains three policies repeated across 17 tasks, giving 51 policy-task cells and reporting  $r = .78, p < .001$  [10]. The policy axis has three blocks and the task axis has 17; each supports a different question and exchangeability argument.

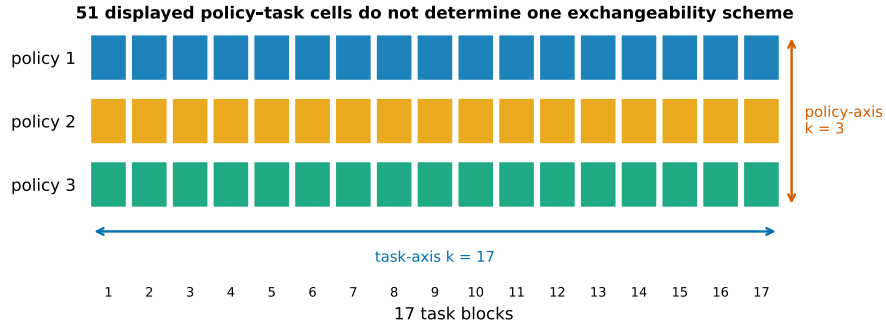


Figure 1: The same 51 cells support distinct policy- and task-axis questions.

The evidence required depends on the claim being made:

| <b>claim</b>                | <b>minimum object</b>                                                                      | <b>what remains unresolved</b>                                     |
|-----------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| Displayed-panel association | Paired real and simulated values, metric definition, and unit identity                     | Which policy an evaluator selects and the real loss of that choice |
| Finite-panel decision       | Candidate set, aggregation and selection rule, ties, real winner set, and loss             | Performance for new policies, tasks, sites, or sessions            |
| Population transport        | Target population plus a sampling, assignment, or predictive model linking it to the panel | Whether the resulting decision is acceptable for the intended use  |
| Decision acceptability      | Named action, loss, tolerance, uncertainty, and downstream real-test budget                | Broader uses not covered by the declared target and tolerance      |

Passing an earlier row does not pass a later one. Conversely, a useful finite-panel result need not make a population claim if its action and boundary are explicit.

### 3. Methods

#### 3.1 Corpus and coding

We included 26 papers that evaluated robot policies, checkpoints, or variants with aligned simulated and real outcomes and printed a named association coefficient. The corpus is non-systematic. We coded each claim’s units, target, transport basis, checkpoint selection, and coefficient uncertainty.

#### 3.2 Numeric data

We extracted numeric results for 19 papers from source data, tables, or figures and checked figure-derived values against reported statistics where possible. Matching a reported coefficient checks extraction, not the underlying experiment. We label figure-derived values and do not infer missing trial counts.

#### 3.3 Decision atlas

For 19 complete matrices, we preserved source-defined rules where available; otherwise we weighted displayed tasks equally, retained ties, and computed displayed real regret. The resulting top-1 rule agrees in 17 cases and disagrees in two.

#### 3.4 Task composition and real-trial remeasurement

We test all displayed-task subsets and leave-one-task-out aggregates; these do not represent future-task probabilities. Where real trial counts are known, independent Beta(1,1) models estimate winner uncertainty. Missing simulator counts and dependence are handled by sensitivity analysis.

## 4. Reporting audit

| result                                           | count                    | interpretation boundary                                            |
|--------------------------------------------------|--------------------------|--------------------------------------------------------------------|
| Included papers                                  | 26                       | non-systematic corpus                                              |
| Complete numeric matrices                        | 19/26                    | source- and figure-derived                                         |
| Finite displayed panel                           | 26/26                    | all audited papers                                                 |
| Defined target population or sampling mechanism  | 0/26                     | none reported                                                      |
| Coefficient uncertainty                          | 5 p; 1 CI; 20<br>neither | p-values test a null rather than quantify<br>magnitude uncertainty |
| Checkpoint or model state selection (yes/no/N/A) | 8 / 17 / 1               | disclosed in eight papers                                          |
| Fewer than ten policy or checkpoint blocks       | 23–25/26                 | range reflects ambiguous reporting                                 |

No paper defines a target population or probability-sampling mechanism, although some make model-based transport arguments. Several papers report held-out prediction, but none defines a future-policy or future-task distribution with calibrated population-level uncertainty. Most reports also do not define the test or exchangeability unit needed for a common inferential interpretation.

## 5. Correlation and displayed decisions

Figure 2 compares illustrative agreement and disagreement cases at nearby, often high correlations. It is not a prevalence sample; the complete 19-case ledger appears in the supplement.

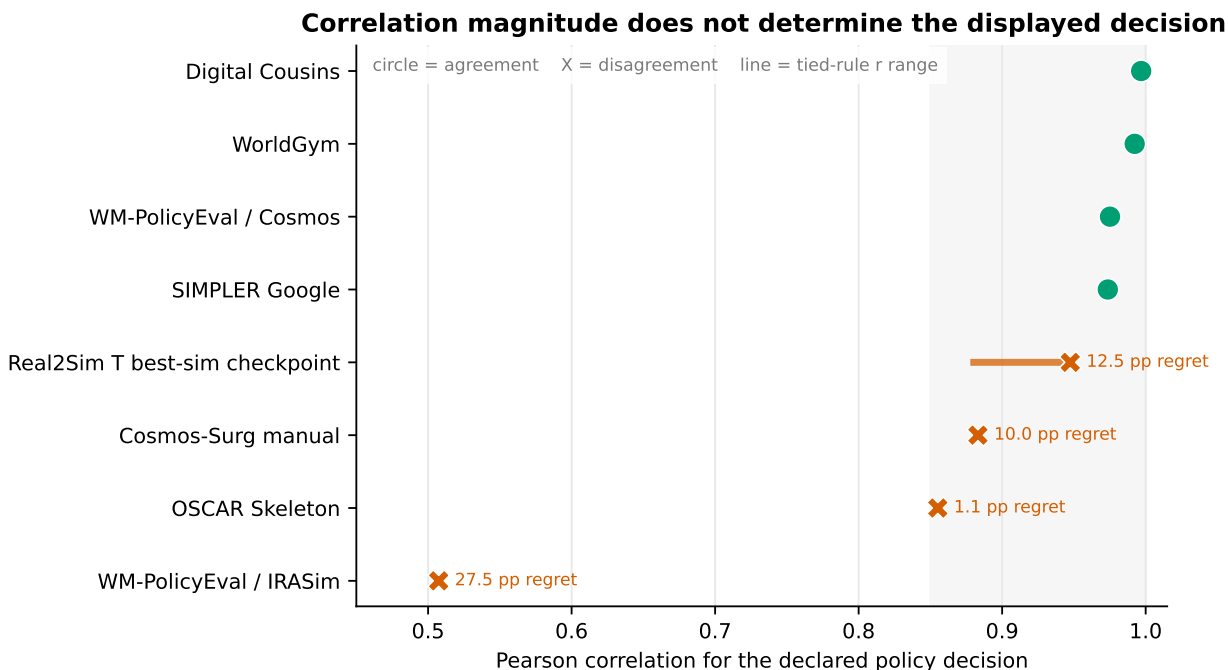


Figure 2: Illustrative cases: similar correlations can yield agreement or disagreement. Labels show displayed real regret. Cases use different units and aggregation rules and are not directly comparable.

| case                     | $r$       | sim best          | real best      | regret  | LOTO  |
|--------------------------|-----------|-------------------|----------------|---------|-------|
| WorldGym                 | .992      | OpenVLA           | OpenVLA        | 0       | 17/17 |
| Digital Cousins          | .997      | $\pi_0$           | $\pi_0$        | 0       | 4/4   |
| SIMPLER Google           | .974      | RT-1-converged    | RT-1-converged | 0       | 5/5   |
| Real2Sim T<br>(best-sim) | .878–.947 | $\pi_0$           | DP             | 12.5 pp | –     |
| OSCAR Skeleton           | .855      | PG-FAST+          | pi0-FAST       | 1.1 pp  | –     |
| Cosmos-Surg<br>manual    | .883      | GR00T N1.5<br>50k | GR00T N1 20k   | 10.0 pp | 0/4   |
| WM-PolicyEval<br>Cosmos  | .975      | OpenVLA           | OpenVLA        | 0       | 4/4   |
| WM-PolicyEval<br>IRASim  | .507      | Octo-Base         | OpenVLA        | 27.5 pp | 1/4   |

LOTO counts robustly correct leave-one-task-out aggregates. WorldGym reports a pooled-cell  $r = .78$ ; equal policy averaging gives  $r = .992$ . For Cosmos-Surg, equal task weighting gives  $r = .883$ ; averaging by training lineage gives  $r = .905$ ,  $\rho = .500$ , and 3.7 points of displayed regret. Real2Sim’s tie changes  $r$  but not the selected winner or regret. Rank-aware metrics reveal additional disagreement but do not define the action, loss, tie rule, or evaluation budget.

Aggregate agreement can conceal task disagreement: SIMPLER Google has  $r = .974$  and an equal-task winner match, but only 3/5 task winners agree. The evaluation rule can also reverse the result: Real2Sim rope moves from  $r = .901$  over all checkpoints to  $r = -.683$  after best-simulated checkpoint selection. Figure 2 shows that nearby coefficients can therefore support different actions.

### 5.1 Agreement cases

WorldGym and Digital Cousins retain the correct winner under every single-task deletion and nearly or fully all task subsets. SIMPLER Google also retains its aggregate winner after every deletion, although only 3/5 task winners agree [6,13]. These examples show stable agreement on their displayed tasks.

### 5.2 Disagreement cases

Real2Sim motivates simulation for checkpoint selection [7]. Selecting each policy’s best displayed simulated checkpoint in the T block chooses  $\pi_0$  instead of the real winner DP, with 12.5 points of displayed regret. An SVLA checkpoint tie changes  $r$  from .878 to .947 but not the decision. Across three tasks and three tie-complete rules, seven of nine disagree, one agrees, and one depends on tie handling.

OSCAR’s plotted values give  $r = .855$  [29]. PG-FAST+ wins on the world-model axis and pi0-FAST on the real axis, a 1.1-point near-tie. The axes use different evaluation pools (65 versus 63 sessions), preventing uncertainty reconstruction. Its printed Spearman  $\rho = .750$  and MMRV = .571 are not jointly reproducible from the displayed bars under the published metric definitions; the required per-policy rank inputs were not released.

Cosmos-Surg’s equal-task aggregate selects GR00T N1.5 50k in simulation and GR00T N1 20k in reality, with 10.0 points of regret [18]. Every single-task deletion retains the mismatch. Real outcomes use ten trials per policy-task, without released dependence information.

WM-PolicyEval changes the decision when only the evaluator changes [31]. Cosmos selects the real winner OpenVLA at  $r = .975$ ; IRASim selects Octo-Base at  $r = .507$ , producing 27.5 points of regret. The plotted points reproduce the appendix values despite an inconsistent IRASim legend, and label-order ambiguity does not change either decision. Simulator counts are unstated.

In *A Practical Recipe*, simulated and real top-1 disagree in 3/11 printed rank panels; absolute regret cannot be recovered [9].

## 6. Robustness to measurement and task composition

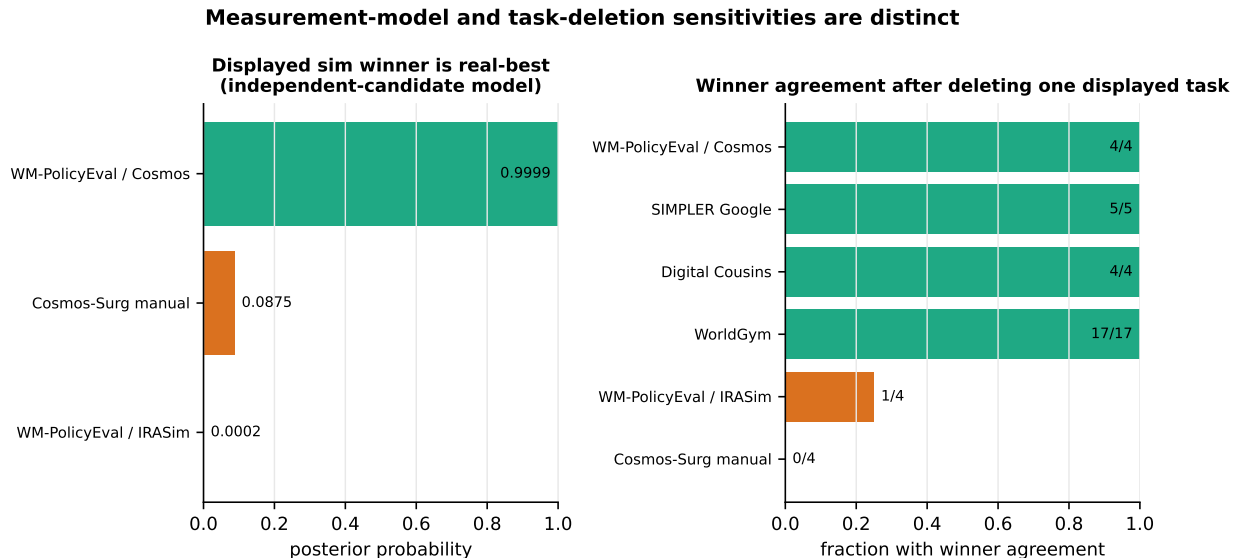


Figure 3: Model and task-composition sensitivities, not calibrated evaluator reliability.

Holding the simulator winner fixed, the Beta-binomial model gives Cosmos-Surg .087 winner concordance and 11.6 points of expected regret. A two-sided sensitivity gives .207–.220 concordance and 8.36–9.65 points of regret; dependence is not identified.

Because WM-PolicyEval omits simulator rollout counts, sensitivity ranges are .333–.9999 for Cosmos and .0001–.432 for IRASim. Policy-specific evidence can reverse IRASim toward OpenVLA; the unknown counts and dependence prevent a reliability estimate.

Both WM evaluators underpredict real success. Positive-slope affine recalibration lowers error without changing rankings. An in-sample isotonic fit changes IRASim’s winner to OpenVLA but uses the same outcomes for fitting and evaluation, so it is only a shape sensitivity.

Task-subset results answer a different robustness question. WorldGym and Digital Cousins remain correct under every leave-one-task-out deletion. Cosmos-Surg manual remains wrong after every deletion. WM-PolicyEval/IRASim is composition-sensitive, becoming robustly correct in one of four leave-one-task-out aggregates. A second deletion is only possibly correct: the simulated winner set ties a real winner with a non-winner. Thus full-panel top-1 alone does not distinguish stable agreement from a composition-dependent result, and tie handling can matter.

## 7. Metric semantics and numerical provenance

Real2Sim’s Figure 3 correlations reproduce within .0004. Treating one coincident rope marker as one or two points changes  $r$  but not any top-1 conclusion.

One tested convention reproduces all six Real2Sim MMRV values: ordering disagreement weighted by the simulated-side gap and divided by  $N$  [6].

## 8. Data and code availability

Data and code are provided with this article. Project materials and checklist: <https://github.com/trilamsr/research/tree/main/sim2real-correlation-audit>.

## 9. Reusable decision audit and reporting standard

The accompanying checklist covers the decision, data, sensitivities, target, loss tolerance, and real-test budget.

## 10. Limitations

This non-systematic 26-paper corpus cannot estimate field prevalence and is concentrated in recent manipulation and world-model evaluation. Figure-derived values remain subject to raster resolution, overlapping marks, source revisions, and unlabeled points despite matching reported coefficients.

The cases differ in robots, tasks, candidates, aggregation, source quality, and loss. They do not identify a universal threshold or population selection-error rate. Real-trial estimates are model-conditional; missing simulator counts and dependence limit their interpretation. Task-subset results describe displayed tasks, not future-task probabilities.

Missing target or uncertainty information limits inference; it does not imply evaluator failure.

## 11. Conclusion

A sim-to-real correlation is a property of a displayed panel. The audited cases show that similar, often high coefficients can accompany different winners and losses under named decision rules. Measurement and task-composition sensitivities then answer distinct questions; neither creates a population claim from a finite panel.

Correlation remains useful evidence of association. It becomes actionable only with a defined target, decision rule, and loss tolerance; estimated outcomes also require decision-relevant uncertainty.

## Author and declarations

Tri Lam conceived the audit, curated the records, implemented the analyses, and wrote the manuscript. No conflicts of interest are declared. The paper is a retrospective public-record audit and reports no new human- or animal-subject experiment.

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